WATCHDOG 2.4

2.4 GHz Wireless Security Analyser

v3.1 ENGINEERING HARDENED| Detect. Classify. Respond.

RF Jammer Detection

Real-time classification of 9 jammer types

WiFi Scanner

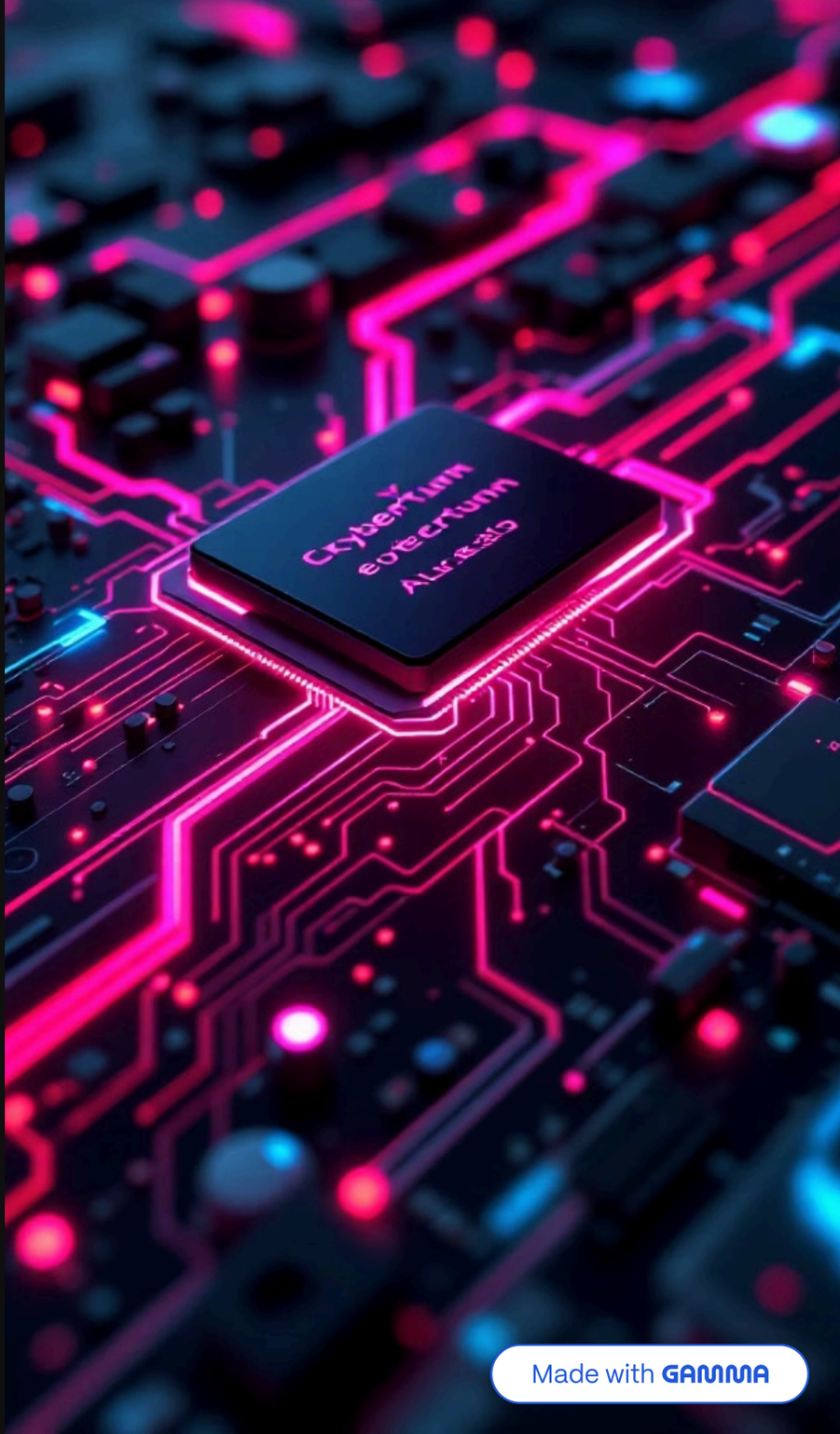
Evil twin detection and network mapping

BLE Mapper

AirTag and SmartTag tracking

Foxhunt Mode

Geiger-counter audio for transmitter tracking



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The Detection Engine

9-Type Jammer Classification System

WATCHDOG 2.4 employs a custom algorithm that continuously scans 125 channels in the 2.4 GHz ISM band, classifying interference patterns with millisecond-level precision. Each detection type triggers immediate alerts while maintaining concurrent WiFi and BLE monitoring.

1

Broadband Burst

80ms detection — Wideband noise floods across multiple channels simultaneously

2

RC Drone Jam

300ms detection — Targets 2.4 GHz RC control links used by consumer drones

3

WiFi Deauth Flood

300ms detection — Deauthentication frame flooding attacks on wireless networks

4

Drone FHSS Link

400ms detection — Frequency-hopping spread spectrum patterns from commercial drones

5

Bluetooth Jammer

400ms detection — Interference targeting Bluetooth audio and HID devices

6

RF Sweep

500ms detection — Sequential channel scanning creating moving interference patterns

7

BLE Jammer

500ms detection — BLE beacon suppression targeting proximity tracking devices

8

WiFi Band Jam

600ms detection — Continuous channel occupation disrupting WiFi communications

9

Anomaly Detection

<700ms detection — Shannon Entropy analysis flags high-entropy anomalous noise floors

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Hardware Architecture

System Block Diagram

01

Core Processor

ESP32 DevKit v1 — 240 MHz dual-core Tensilica LX6 running FreeRTOS

02

RF Front-End

nRF24L01+ PA/LNA Module — 125-channel scanning with -64 dBm sensitivity

03

Display System

ILI9341 320×240 TFT with XPT2046 resistive touch — Shared SPI bus architecture

04

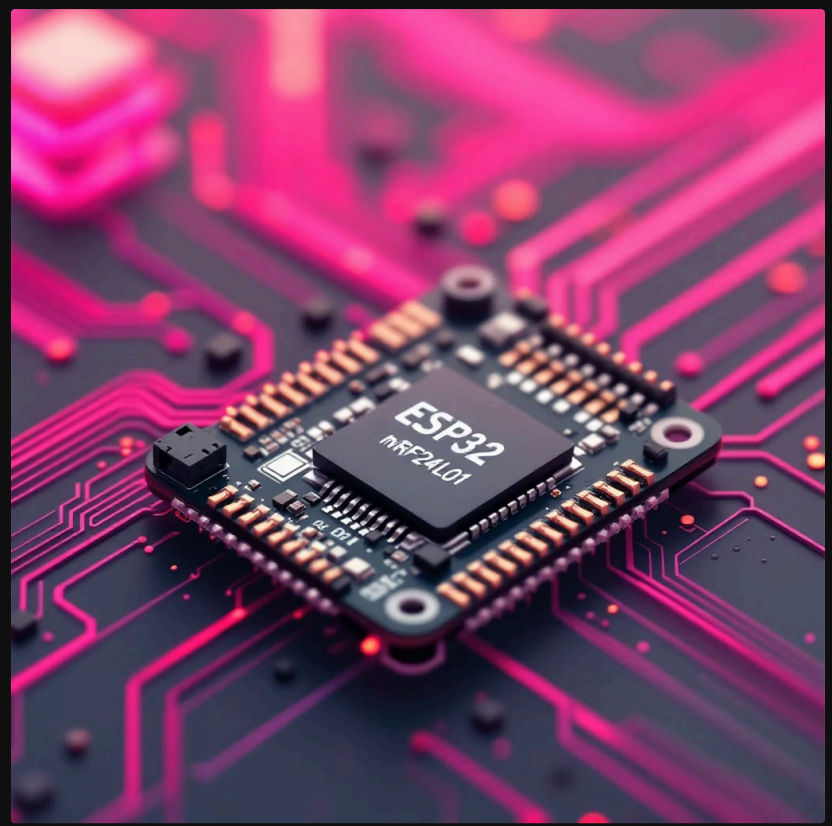
Power Management

USB-C to TP4056 LiPo charge IC → 5V LDO → Module power distribution

05

Audio Alerts

Active 3.3V PWM Buzzer — Geiger-counter tone with HOT/COLD feedback





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Tactical Capabilities

Feature Modules



WiFi Scanner

Asynchronous channel hopping detects evil twins, camera OUIs, and channel congestion patterns. Identifies unauthorized access points and rogue devices operating on common IoT frequencies.



BLE Mapper

Continuous Bluetooth Low Energy monitoring tracks AirTags, Apple Find My network devices, Samsung SmartTags, and hidden MAC addresses. Proximity tracking enables real-time presence detection.



RF Foxhunt

Geiger-counter audio feedback with dynamic HOT/COLD indicators creates intuitive physical transmitter tracking capability. Adjusts detection sensitivity based on movement velocity for optimal direction finding.



Wireless IDS

Live alert log with 5 persistence counter gauges tracks attack progression across multiple vectors. Correlates events across jammer types, WiFi anomalies, and BLE patterns for comprehensive threat assessment.



Power Management

ECO Mode delivers 10-hour runtime on 3.3V 300 mAh cell. **Auto Sleep** achieves 10 µA deep sleep current for extended deployment. USB-C charging supports 5V/2A input with thermal protection.



Passive Sensing

Zero transmission policy ensures complete RF stealth. All detection operates through receive-only mode using nRF24L01+ PA/LNA hardware. No packet injection or active probing capabilities.

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Touchscreen Interface

6-Page UI Showcase

1	**HOME Screen** Critical alerts summary with priority indicators and system status overview
2	**RF SPECTRUM** Live 125-channel waterfall display with color-coded power levels and jammer classification
3	**WiFi SCANNER** Network discovery with SSID, BSSID, channel, and security protocol analysis
4	**BT MAPPER** Bluetooth device proximity tracking with MAC anonymization and device type identification
5	**WIRELESS IDS** Real-time intrusion detection with attack timeline and persistence counter visualization
6	**SYSTEM STATUS** Resource monitoring showing CPU usage, memory allocation, battery level, and uptime

800ms Long-Press Sub-Modes

Stealth Mode (UI dimming), Focus (single-channel monitoring), Freeze (pauses scanning), ChanMap (channel allocation visualization)

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Tech Specs & Architecture

Complete Specifications & Stability

750

Channels Per Second

6 channels processed every 10 ms tick across 125-channel scan range

-64

dBm Sensitivity

Baseline trigger threshold for jammer detection and anomaly classification

20

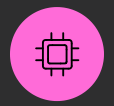
Hour Runtime

ECO Mode operation on 300mAh 3.3V cell with USB-C charging support

10

µA Deep Sleep

Auto Sleep power conservation for extended deployment scenarios



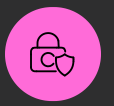
FreeRTOS Architecture

Single-loop cooperative non-blocking design with delay(1) watchdog feed guarantees zero crashing under normal operation



Firmware Stack

Custom detection algorithms written in C++ with platform-specific optimizations for ESP32 and nRF24L01+ hardware



Passive RF Sensing

Receive-only operation ensures complete RF stealth with no transmission capabilities or active probing features

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Created by **Anirudh N.S.** | Passive RF Sensing Only | Personal Cybersecurity Portfolio Project

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